

Python Exception Handling and File Operations

Python uses exceptions to handle errors gracefully, preventing program crashes from unexpected conditions. The try-except-else-finally structure provides comprehensive error handling. Try blocks contain code that might raise exceptions, except blocks catch specific exceptions and provide recovery logic, else blocks execute when no exception occurs, and finally blocks guarantee execution regardless of exceptions. Multiple except clauses handle different exception types. Raising exceptions allows explicit error signaling using the raise keyword. Custom exception classes inherit from built-in exceptions for application-specific errors. File operations in Python are straightforward using the open() function with context managers (with statement) ensuring proper resource cleanup. Files can be opened in various modes: 'r' for reading, 'w' for writing, 'a' for appending. Reading methods include read() for entire content, readline() for single lines, and readlines() for all lines. Writing methods include write() and writelines(). The with statement ensures files are automatically closed after use. Python's pathlib module provides object-oriented file path handling for cross-platform compatibility.